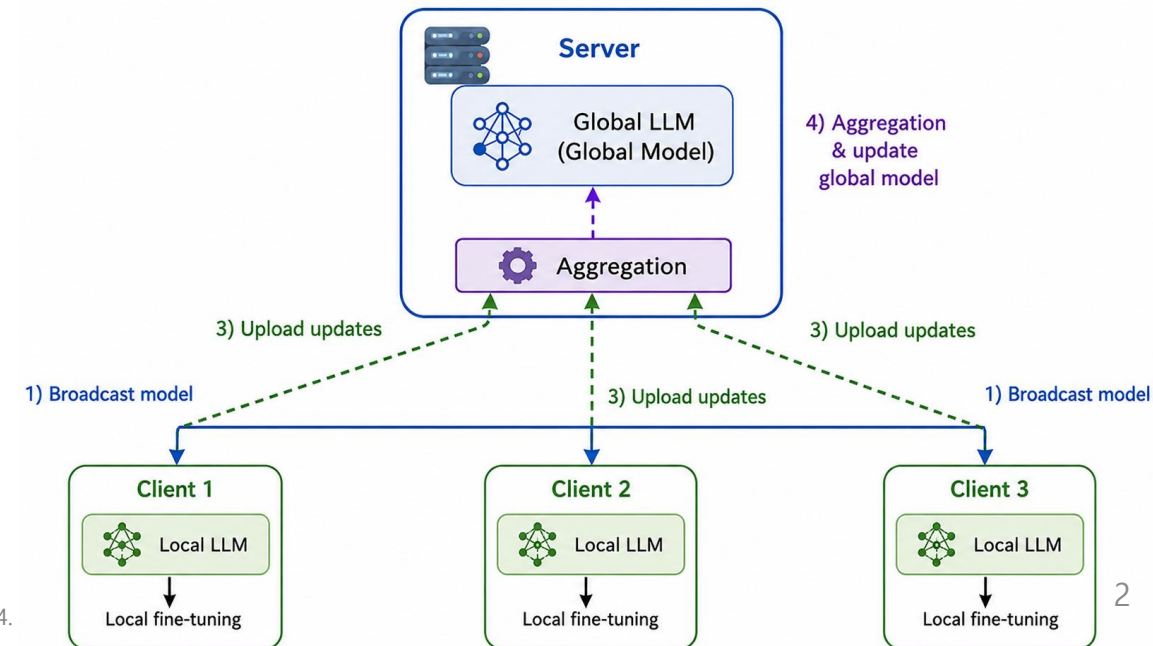


Test-Time Adaptation for Robust Personalized Federated LLMs in the Medical Domain

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Introduction

- LLM을 의료 도메인 downstream task에 적용하려면 대규모 fine-tuning이 필요하지만, **의료 데이터는 병원별로 분산되어 있고 민감정보라 직접 공유가 어려움.** [1][2]
- **Federated Learning(FL)**은 원시 데이터를 공유하지 않고도 협력 학습이 가능해, 이러한 프라이버시 제약의 대안이 됨. [2][3]
- 다만 LLM의 방대한 파라미터는 통신 병목을 유발하므로, **PEFT(LoRA) 기반 연합학습**으로 발전하고 있음. [4][5]
- 또한 병원·진료과마다 데이터 분포가 이질적이어서 단일 글로벌 모델로는 한계가 있어, **개인화 Personalized FL**이 필요. [6]



[1] Singhal, K. et al. Large Language Models Encode Clinical Knowledge. Nature, 2023.
[2] Rieke, N. et al. The Future of Digital Health with Federated Learning. npj Digital Medicine, 2020.
[3] McMahan, B. et al. Communication-Efficient Learning of Deep Networks from Decentralized Data. AISTATS, 2017.
[4] Bian, J. et al. A Survey on Parameter-Efficient Fine-Tuning for Foundation Models in Federated Learning. arXiv, 2025.
[5] Wang, Z. et al. FLoRA: Federated Fine-Tuning Large Language Models with Heterogeneous Low-Rank Adaptations. NeurIPS, 2024.
[6] Tan, A. Z. et al. Towards Personalized Federated Learning. IEEE TNNLS, 2022.

Introduction

- 실제 임상 환경에서는 학습 때 보지 못한 Query(타 과 환자·희귀질환 등)가 유입되어 **Test-time distribution shift**가 발생. [7]
- 그러나 기존 **Personalized FL**은 로컬 분포에만 최적화되어 있어 이러한 분포 변화에 취약. [7]
- 특히 의료 분야에서 오답은 오진·환자 안전 문제로 직결되므로, 분포 변화에 대한 robustness는 임상 신뢰성 요건. [8]
- 이에 본 연구는 Test-time distribution shift 문제를 해결하기 위해, **inference 시점에 Entropy Minimization 기반 Test-Time Adaptation(TTA)**을 통해 분포 변화에 적응하는 방법 제안. [9]

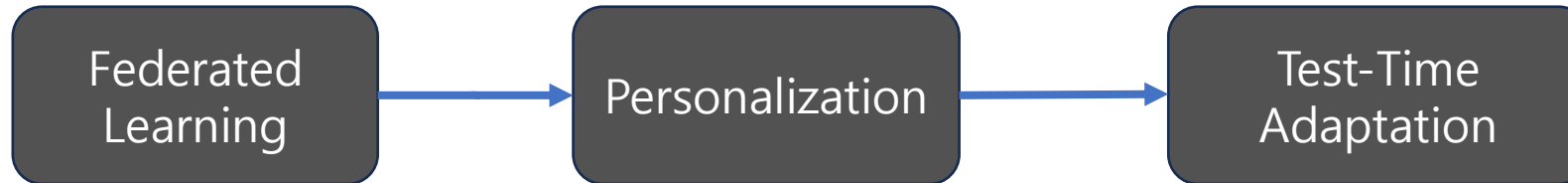
[7] Yang, Y. et al. Dual-Personalizing Adapter for Federated Foundation Models (FedDPA). NeurIPS, 2024.

[8] Finlayson, S. G. et al. The Clinician and Dataset Shift in Artificial Intelligence. NEJM, 2021.

[9] Wang, D. et al. Tent: Fully Test-Time Adaptation by Entropy Minimization. ICLR, 2021.

Objective

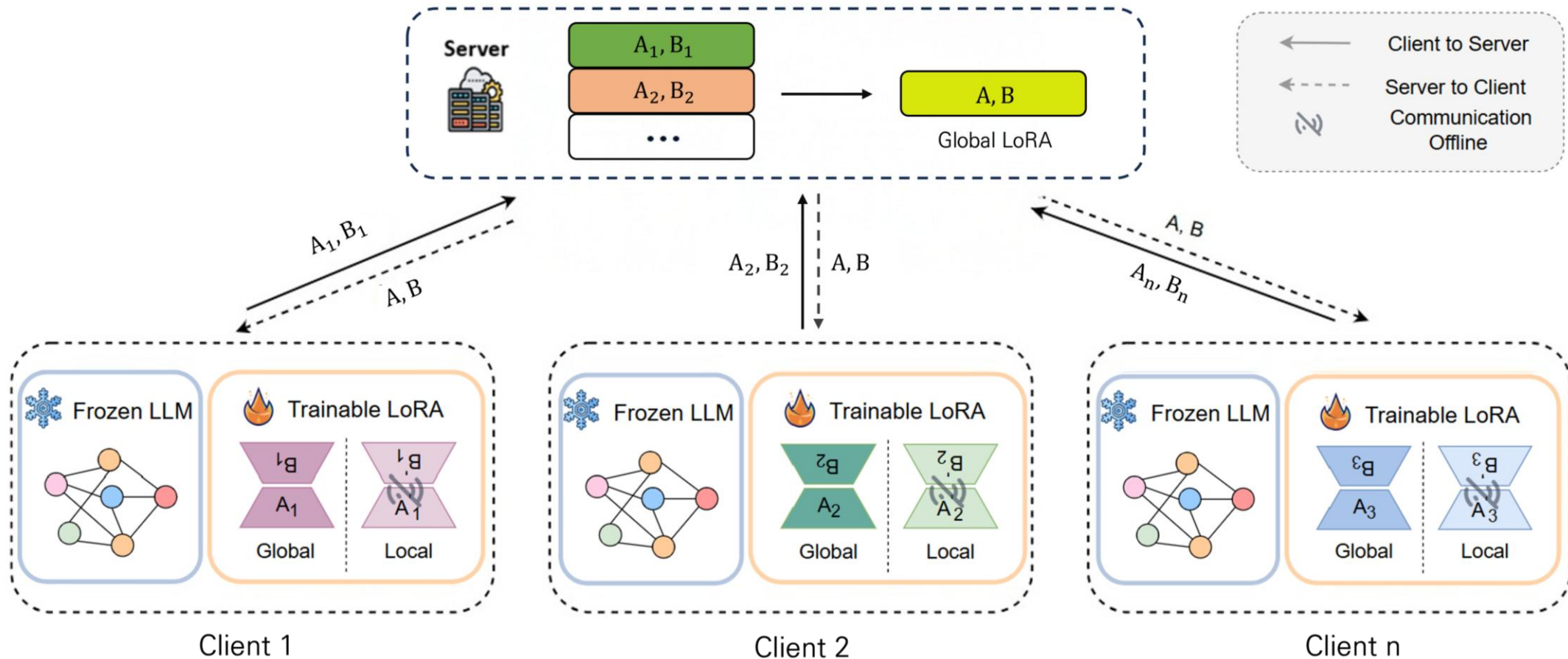
- 도메인이 이질적인 의료 FL 환경에서, 로컬 도메인의 **Personalization** 성능을 유지하면서
학습 시 보지 못한 Query (unseen query)로 인한 **Test-time distribution shift** 상황에도 **robust**하게 대응



Overall Framework

Training Phase: Federated Learning

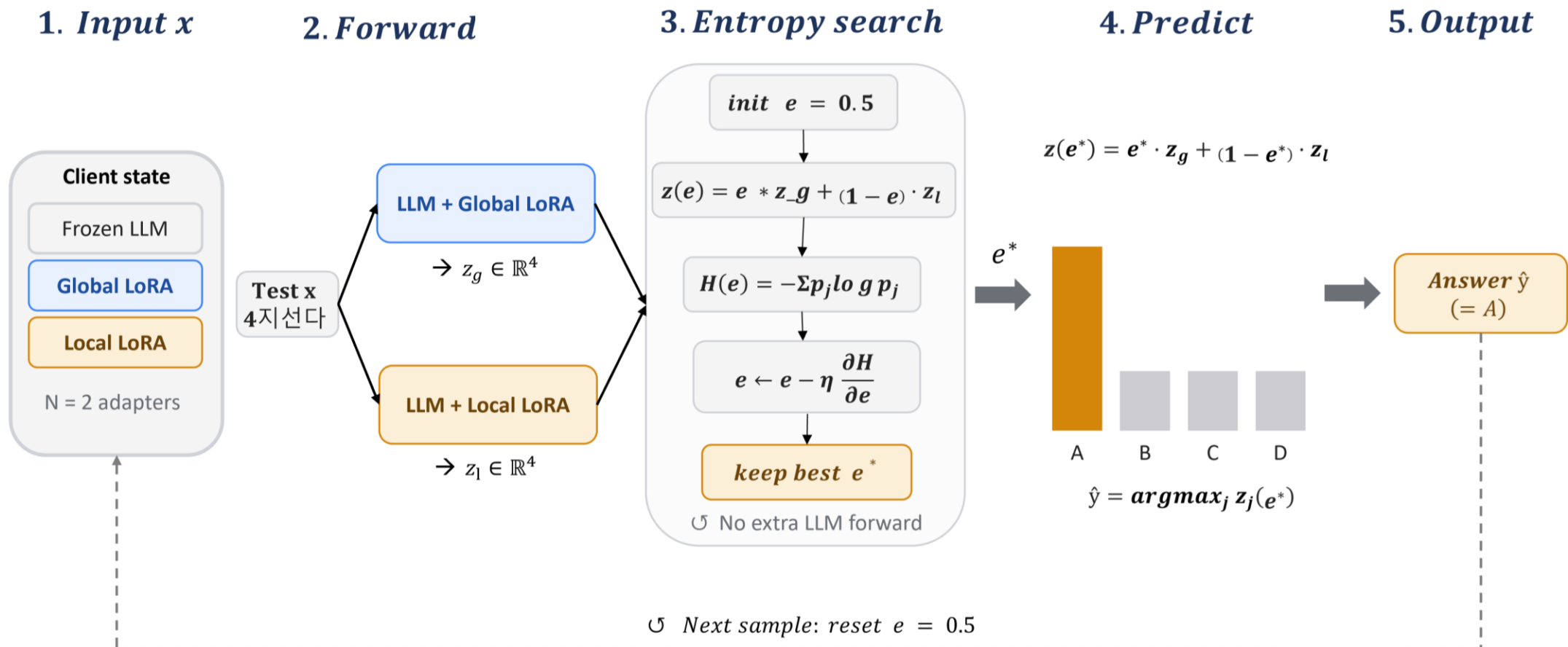
- ✓ Frozen LLM 위에 클라이언트마다 **Global · Local LoRA** 학습



Overall Framework

Inference Phase: (Entropy Minimization Test-time Adaptation)

- ✓ Test-time에 test query별로 4지선다 분포의 **entropy**가 최소가 되도록 2개의 LoRA의 결합 가중치 e 를 조정



Experiment Setting

- **Total Loss:** LM task loss
- **Model Backbone**
 - ✓ Qwen3-4B-Base
- **Configuration**
 - ✓ Communication Rounds, Epochs: **10 rounds, 3 epochs**
 - ✓ Optimizer: AdamW
 - ✓ LoRA
 - ✓ **Rank = 8 , Q,V projection**
- **결합 가중치 update setting**
 - ✓ Steps: 20
 - ✓ Learning rate: 0.1

Dataset (MedMCQA)

- MedMCQA는 인도 의대 입학시험 기출 기반의 약 194k 문항 • **21개 의료 과목 4지선다** 의료 QA 벤치마크 데이터셋
- **4 Domain = 4 client** → 진료과를 이질적 도메인으로 매핑(내과, 피부과, 외과, 정형외과)
- 정답 letter 균등화(A/B/C/D 25%)를 통해 위치 편향 제거
- Distribution shift robustness 평가용 **800개**(4 domain 혼합)

Client	Domain	Train	Test
1	Medicine(내과)	600	200
2	Skin(피부과)	600	200
3	Surgery(외과)	600	200
4	Orthopaedics(정형외과)	600	200

Results —(Accuracy)

- **Personalization:** 로컬 test 데이터셋 200개 평가.
- **Test-time Generalization:** 전체 4개의 test 데이터셋 800개(200 samples x 4)을 활용하여 Robustness 평가.
- **Metric** = Accuracy

Method	Medicine	Skin	Surgery	Orthopaedics	Average
Personalization					
Base(frozen)	42.0	50.5	36.0	36.5	41.2
Local-finetuning	62.5	69.5	61.0	62.0	63.8
FedIT	63.0	70.5	62.5	63.5	64.9
FedDPA(NeurlIPS,2024)	62.5	72.0	63.5	67.0	66.3
Ours	62.5	73.5	64.0	68.0	67.0
Test-time Generalization					
Base	41.2	41.2	41.2	41.2	41.2
Local-finetuning	61.8	62.5	62.1	60.6	61.8
FedDPA(NeurlIPS,2024)	65.9	65.0	65.3	66.0	65.5
Ours	65.9	65.4	66.5	67.3	66.3

Conclusion & Limitations

Conclusion

- inference 시점에 **2개의 LoRA를 query별로 동적으로 결합**하는 Entropy-Minimization TTA 방법론 제안.
- 추가적인 모델 학습 및 통신 없이 Test-time Distribution shift 상황에 **robust한 개인화**를 구현.
- 학습 시 보지 못한 Query는 **Global LoRA**, 본인 도메인 Query는 **Local LoRA**로 동적으로 **배정**되어 사전연구들 대비 Unseen query 성능 향상.

Limitation

- Inference 시점의 LoRA 결합 가중치 조정하여 고정된 LoRA의 표현 범위에 성능 제한, 향후 **학습 단계의 개인화 강화 방안 보완 필요.**
- 4개 진료과가 일반 의학지식을 공유하여 분포 변화가 약하고 entropy(확신)가 정답과 항상 일치하지 않아, 향후 **이질적인 도메인 구성 및 3 seed 실험을 통해 추가 검증 필요.**



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Query에 따른 가중치 반영 실제 결과 예시

- Client 1(Medicine)

질문 + 보기	global	local	예측
[Orthopaedics] Q. Spring ligament refers to - (A) Sho plantar ligament (B) Deltoid ligament (C) Plantar calcaneo navicular ligament (D) Long plantar ligament	0.95	0.05	C ✓
[Orthopaedics] Q. Judet and LECTOURNAL classification is used for (A) Shaft femur fracture (B) Acetabulum fracture (C) Patella fracture (D) Supracondylar Humerus fracture	0.90	0.10	B ✓
[Orthopaedics] Q. Most common malignant bone tumor is: (A) Chondrosarcoma (B) Osteosarcoma (C) Metastasis (D) Ewing's sarcoma	0.88	0.12	C ✓

질문 + 보기	global	local	예측
[Medicine] Q. Infliximab is contraindicated in (A) Crohn's disease (B) Rheumatoid arthritis (C) Intestinal tuberculosis (D) Ankylosing spondylitis	0.06	0.94	C ✓
[Medicine] Q. Hepatocellular carcinoma is associated with infection caused by - (A) Hepatitis G virus (B) Hepatitis E virus (C) Hepatitis A virus (D) Hepatitis C virus	0.06	0.94	D ✓
[Medicine] Q. Which of the following is not a side effect of blood transfusion: September 2009 (A) Hypokalemia (B) Iron overload (C) Hypomagnesemia (D) Hypocalcemia	0.06	0.94	A ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 2(Skin) – 외부 Query

질문 + 보기	global	local	예측
[Surgery] Q. A 49-year-old man suffering from depression attempts suicide by jumping out of the window of his third-floor apartment. He requires multiple operations during a prolonged, complicated hospital stay. Endotracheal intubation is attempted in the ICU but is unsuccessful because of tracheal stenosis, which is attributed to which of the following? (A) Prolonged nasotracheal intubation (B) Tracheostomy tubes (C) High oxygen delivery (D) Orotracheal intubation	0.95	0.05	A ✓
[Orthopaedics] Q. Kinebock's disease is due to avascular necrosis of- (A) Femoral neck (B) Lunate bone (C) Medial cuneiform bone (D) Scaphoid bone	0.94	0.06	B ✓
[Surgery] Q. In follow up of BPH, most important indication of surgery is: (A) Prostate size >75gm (B) Single episode of UTI requiring 3 day antibiotics (C) Cannot use medication due to hypertension (D) Bilateral hydronephrosis	0.94	0.06	D ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 2(Skin) – Local Query

질문 + 보기	global	local	예측
[Skin] Q. Acrodermatitis entero pathica is d/t deficiency of: (A) Se (B) Zn (C) Cu (D) Cr	0.08	0.92	B ✓
[Skin] Q. A 10-year-old boy presented with painful boggy swelling of scalp, multiple sinuses, with purulent discharge, easily pluckable hair and lymph nodes enlarged in occipital region. Which will be most helpful for diagnostic evaluation? (A) Biopsy and Giemsa evaluation (B) Patch test, gram staining, Tzanck smear (C) KOH mount (D) Bacterial culture	0.18	0.82	C ✓
[Skin] Q. Bubos with multiple sinuses discharging into inguinal lymph nodes are seen in: (A) Chancroid (B) Syphilis (C) LGV (D) Granuloma Inguinale	0.27	0.73	C ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 3(Surgery) – 외부 Query

질문 + 보기	global	local	예측
[Medicine] Q. Low QRS voltage on ECG indicates - (A) Pericardial effusion (B) Pulmonary embolism (C) Infective endocarditis (D) Cor pulmonale	0.94	0.06	A ✓
[Orthopaedics] Q. Gradual painful limitation of shoulder movements in an elderly suggest that the most probable diagnosis is: (A) Ahritis (B) Osteoahritis (C) Myositis ossificans (D) Periahritis	0.94	0.06	D ✓
[Orthopaedics] Q. Main blood supply to the head and neck of femur comes from (A) Medial circumflex femoral Aery (B) Popliteal Aery (C) Lateral circumflex femoral Aery (D) Aery of Ligamentum Teres	0.93	0.07	A ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 3(Surgery) – Local Query

질문 + 보기	global	local	예측
[Surgery] Q. A 3 week old child with projectile postprandial vomiting is diagnosed of having hyperophic pyloric stenosis. Congenital hyperophic pyloric stenosis is associated with: (A) Hypokalemic alkalosis (B) Hypokalemic acidosis (C) Hyperkalemic alkalosis (D) Hyperkalemic acidosis	0.05	0.95	A ✓
[Surgery] Q. Abbe - Estlander is used for: (A) Ears (B) Tongue (C) Lip (D) Eyelid	0.07	0.93	C ✓
[Surgery] Q. The treatment of post traumatic epilepsy is - (A) Long term anticonvulsants (B) Long term corticosteroids (C) Immediate corticosteroids (D) Mannitol infusion	0.08	0.92	A ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 4(Orthopaedics 모델) – 외부 Query

질문 + 보기	global	local	예측
[Surgery] Q. In intestinal anastomosis, strength is mainly provided by:- (A) Mucosa (B) Submucosa (C) Muscularis mucosa (D) Serosa	0.95	0.05	B ✓
[Surgery] Q. Maximum dose of xylocaine with out adrenaline that can be given in 60kg adults is (A) 600mg (B) 300mg (C) 400mg (D) 500mg	0.93	0.07	B ✓
[Skin] Q. Dennie Morgan folds are seen in (A) Nummular eczema (B) Keratosis pilaris (C) Lichen planus (D) Atopic dermatitis	0.92	0.08	D ✓

Query에 따른 가중치 반영 실제 결과 예시

- Client 4(Orthopaedics 모델) – 내부 Query

질문 + 보기	global	local	예측
[Orthopaedics] Q. Housemaids knee is bursitis of - (A) Prepatellar bursa (B) Olecranon (C) Infrapatellar bursa (D) Ischial bursa	0.05	0.95	A ✓
[Orthopaedics] Q. Marble bone disease is (A) Osteopetrosis (B) Osteogenesis imperfect (C) Osteoporosis (D) Osteomyelitis	0.08	0.92	A ✓
[Orthopaedics] Q. Non-Inflammatory arthritis is (A) Reiter's syndrome (B) Gonococcal arthritis (C) Osteo-arthritis (D) Rheumatoid arthritis	0.08	0.92	C ✓

client×도메인 4×4 — Local-finetuning

- Test-time 시기에 각 도메인의 test dataset을 활용한 Local 및 unseen 도메인에 대한 성능

모델 \ test 도메인	Medicine	Skin	Surgery	Ortho
C1·Medicine model	62.5	66.0	57.5	61.0
C2·Skin model	63.0	69.5	59.5	58.0
C3·Surgery model	63.0	67.5	61.0	57.0
C4·Ortho model	61.0	60.0	59.5	62.0

Own 평균 = 63.8, Foreign 평균 = 61.1 → gap = +2.7
 대각행렬(own) 우위

client×도메인 4×4 — Ours

- Test-time 시기에 각 도메인의 test dataset을 활용한 Local 및 unseen 도메인에 대한 성능

모델 \ test 도메인	Medicine	Skin	Surgery	Ortho
C1 · Medicine model	62.5	73.0	63.0	65.0
C2 · Skin model	61.5	73.5	62.0	64.5
C3 · Surgery model	62.0	74.0	64.0	66.0
C4 · Ortho model	64.0	73.5	63.5	68.0

Own 평균 = 67.0, Foreign 평균 = 66.0 → gap = +1.0
 Entropy 라우팅을 통해 Local & Global LoRA 잘 배정